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Transplant-ready? Evaluating AI lung segmentation models in candidates with severe lung disease

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ABSTRACT. **Purpose:** This study evaluates publicly available deep-learning-based lung segmentation models in transplant-eligible patients to determine their performance across disease severity levels, pathology categories, and lung sides, and to identify limitations impacting their use in preoperative planning in lung transplantation.

Approach: This retrospective study included 32 patients who underwent chest CT scans at Duke University Health System between 2017 and 2019 (a total of 3645 2D axial slices). Patients with standard axial CT scans were selected based on the presence of two or more lung pathologies of varying severity. Lung segmentation was performed using three previously developed deep learning models: Unet-R231, TotalSegmentator, and MedSAM. Performance was assessed using quantitative metrics (volumetric similarity, Dice similarity coefficient, and Hausdorff distance) and a qualitative measure (four-point clinical acceptability scale).

Results: Unet-R231 consistently outperformed TotalSegmentator and MedSAM in general for different severity levels and pathology categories ($p < 0.05$). All models showed significant performance declines from mild to moderate-to-severe cases, particularly in volumetric similarity ($p < 0.05$), without significant differences among lung sides or pathology types.

Conclusions: Unet-R231 provided the most accurate automated lung segmentation among evaluated models, with TotalSegmentator being a close second, although their performance declined significantly in moderate-to-severe cases, emphasizing the need for specialized model fine-tuning in severe pathology contexts.

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1 Introduction

Donor–recipient size matching is crucial for successful lung transplantation to ensure the donor lung fits within a fixed thoracic cavity. The current lung allocation system incorporates recipient height as a surrogate for thoracic cavity size to improve size matching[1]; however, this approach remains imprecise and disproportionately affects shorter candidates. Traditionally, predicted total

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lung capacity (pTLC), derived from height-based equations,[2] serves as a proxy for alveolar space. Yet, pTLC fails to account for pathologic alterations of the thoracic cavitysuch as hyperinflation, fibrosis, or effusionswhich frequently occur in patients with end-stage lung disease and often distort the anatomical confines of the thoracic cavity. Because size matching directly influences both graft selection and postoperative outcomes,[3] there is increasing demand for more precise, anatomy-based metrics.

Computed tomography (CT) volumetry has emerged as a reliable and practical solution for preoperative thoracic cavity assessment, offering direct measurements that are unavailable through spirometry or anthropometry.[4] Furthermore, CT is routinely obtained during the transplant evaluation process, enabling opportunistic use for volumetric analysis without requiring additional imaging or testing.[5] For single lung transplants, CT allows for targeted assessment of the transplant hemithorax, critical in patients with asymmetric pathology or prior surgical interventions. In addition, prior work has demonstrated that CT-derived lung volumes correlate more strongly with actual TLC than pTLC and outperform pTLC in predicting posttransplant survival.[4,6] However, accurate CT volumetry hinges on proper segmentation of thoracic structures. Manual segmentation remains the gold standard but is time-consuming, labor-intensive, operator-dependent, and infeasible at scale.[7] Automated segmentation, particularly deep learning-based models, offers a faster and more reproducible alternative. Publicly available models such as Unet-R231,[8] TotalSegmentator,[9] and MedSAM[10]–have demonstrated strong performance in lung CT analysis, but their application in LTx remains limited and largely invalidated.

Despite these advantages, applying automated segmentation models in LTx recipients presents several challenges. First, deep learning-based models often struggle in the setting of severe pathology—such as dense consolidations, emphysema, fibrosis, and pneumothorax—which are common among LTx candidates. Model performance depends more heavily on the diversity and quality of training datasets than on model architecture.[8] However, most publicly available algorithms were developed using datasets containing normal or mildly diseased lungs. Second, segmentation boundaries remain inconsistently defined: some models exclude collapsed or nonaerated lung regions, fail to differentiate solid versus ground-glass opacities, or ignore the presence of pleural air and fluideach of which impacts total volume calculations and leads to inaccurate size matching. These challenges highlight the need for rigorous validation of publicly available automated lung segmentation models to assess their reliability in LTx.

Accordingly, this study evaluates the performance of publicly available automated segmentation models in transplant-eligible candidates, considering different lung sides, disease categories, and severity strata. We hypothesize that current models, while promising in more general populations, will demonstrate variable performance in the pathologically distorted lungs of LTx candidates, limiting their immediate clinical applicability. We assess each model's accuracy and clinical interpretability, as well as key limitations that may impact preoperative planning. By identifying model limitations, this study offers insights into improving segmentation methods for LTx applications.

2 Methods and Materials

This retrospective study was approved by the Duke University Institutional Review Board (Pro00106785) with a waiver of informed consent. The study followed the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) guidelines.

2.1 Dataset

This study was designed to evaluate the performance of deep-learning-based lung segmentation models across a wide range of lung pathologies with varying severity.

The dataset was derived from patients who underwent chest CT scans at Duke Health System between 2016 and 2020. Exams were deidentified, and we used an honest broker service within Duke's secure Protected Analytics Computing Environment (PACE). Standard axial chest CT scans were chosen for their high anatomical resolution and widespread clinical use in lung assessment. The Series Description field from the DICOM header was used to identify the appropriate imaging sequence. All CT scans in this cohort were acquired during inspiration. If multiple standard axial sequences were available for a patient, the scan with the larger slice thickness was

selected to reduce annotation burden. Patients were included based on the presence of the following lung pathologies in radiology reports: atelectasis, consolidation, ground-glass opacities, interlobular septal thickening, nodules, masses, emphysema, fibrosis (reticular, bronchiectasis, distortion, and ground-glass), pleural effusion, pneumothorax, and pleural thickening/nodularity. These lung pathologies were chosen for their clinical significance, prevalence in LTx scenarios, and diagnostic feasibility on CTs. The severity of each lung abnormality was categorized as either mild (encompassing mild, small, or few abnormalities) or moderate-to-severe (moderate-to-severe, medium-to-large, or multiple-to-many) by M.R.H. (a fellowship-trained cardiothoracic radiologist with 2 years of postfellowship experience). A patient was classified as moderate-to-severe if any of their lung abnormalities were assessed as moderate-to-severe. Two sampling strategies were used: (1) random selection of mild cases ($n = 10$) and (2) manual selection of moderate-to-severe cases ($n = 22$). The final dataset included 32 standard axial chest CT scans (3645 2D axial slices) in NIfTI format from 32 patients with two or more of the 11 lung pathologies. The inclusion/exclusion criteria and evaluation pipeline are outlined in Figs. 1 and 2.

2.2 Annotation

In this study, lungs were defined as the entire pleural space, including lung tissue and any associated fluid or abnormalities, while excluding extrathoracic structures such as mediastinum, chest wall, and external air pockets. The reference standard was set by the consensus of M.G.H. (around 25 years of experience in LTx) and M.R.H. Initial annotations were manually annotated by J.L. (no formal radiology training) using 3D Slice,[11] and then were reviewed and approved by M.R.H. to ensure accuracy. Left and right lungs were individually labeled. All cases were reviewed by experts, and annotations were corrected when needed, including both structural corrections and minor boundary refinements. Initial annotation required up to ~3 h per case.

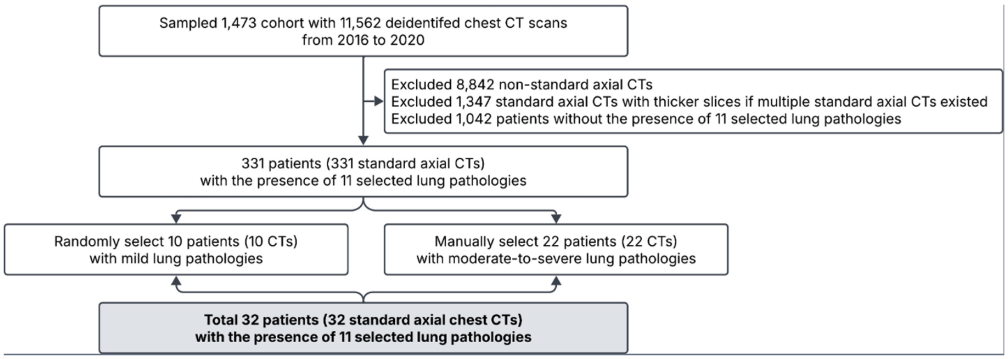


Fig. 1 Inclusion/exclusion criteria.

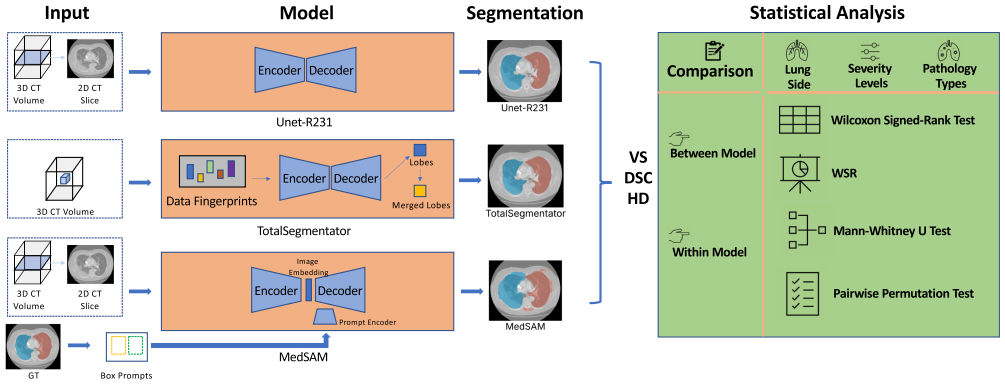


Fig. 2 Experiment pipeline. U-Net-R231, TotalSegmentator, and MedSAM were evaluated on 3D CT volumes using volumetric similarity (versus), Dice similarity coefficient (DSC), and Hausdorff distance (HD), followed by subgroup statistical analysis.

Expert review and revision took up to ~2 h per round and were repeated for additional rounds until the annotations were approved. Quality control was performed by direct visual assessment with iterative correction.

2.3 Deep-Learning-Based Lung Segmentation Models

Three publicly available deep-learning-based lung segmentation models were chosen for their popularity and relevance to lung segmentation tasks. Detailed characteristics of these models are presented in Table 1.

Unet-R231 is a classic 2D Unet model trained on 231 chest CTs from clinical routine, involving 55 cases with severe lung pathologies, such as consolidations, effusions, fibrosis, ground-glass opacity, tumors, and trauma. It processed 3D images as a series of 2D slices and segmented left and right lungs separately. We used the open-source implementation from lungmask GitHub and selected the pretrained Unet-R231 model (listed as the first model in GitHub documentation).

Table 1 Details of three publicly available lung segmentation models.

Feature	Unet-R231	TotalSegmentator	MedSAM
Backbone model	U-Net	nnU-Net	SAM
Target task	2D lung CT segmentation	3D whole-body CT segmentation	2D multimodal medical image segmentation
Training dataset*	Chest CTs from clinical routine practice	Whole-body CTs from clinical routine practice	Publicly available medical image datasets
# of patients	231	1082	Not provided
# of CT scans	231	1082	6226
# of CT scans presenting lungs	231	Not provided	Maximum 1564**
# of CT scans presenting lung pathologies	Minimum 55	Not provided	Not provided
Lung definition	Left and right lungs individually	Left and right lungs individually, combining lobes for each side	Bounding-box prompts based on lung masks from public datasets
Exclusion	Trachea	Trachea***	Not specified
Inclusion	Air pockets, tumor, and pleural effusions	Air pockets, tumor, and pleural effusions***	Not specified
Lung pathologies	Severe pathologies (e.g., consolidations, effusions, fibrosis, ground-glass opacity, tumors, and trauma)	No specific lung abnormalities; includes five broad abnormal categories (bleeding, inflammation, tumor, and vascular)	No specific lung abnormalities
Annotation method for lungs	Manual refinement of auto-segmentations using U-Net trained on VISCERAL Anatomy3 dataset (18)	Manual refinement of auto-segmentations using U-Net–Lobe (19)	Masks from publicly available datasets
Codes available	github.com/JoHof/lungmask	github.com/wasserth/TotalSegmentator	github.com/bowang-lab/MedSAM.

*Training dataset only, not including validation or test datasets.
**Estimated from the combination of training CT datasets where segmentation targets are whole-body or abdominal organs and lungs.
***Not specified in the paper; information derived from the source paper of the model used for annotation.

TotalSegmentator (version 2) is a 3D nnUnet model, an advanced Unet-based architecture that automatically configures hyperparameters based on dataset characteristics.[9] It was trained on 1082 whole-body CTs, part of which contained lungs. Although its training dataset did not specifically focus on lung abnormalities, it included various abnormalities. TotalSegmentator initially segmented lobes for each side and subsequently merged them to generate left and right lung masks.

MedSAM is a fine-tuned Segment Anything Model (SAM)[12] based on a 2D vision transformer model utilizing user-defined box prompts to guide segmentationfor multimodal medical imaging. It was trained on diverse public medical image datasets, including 6226 CTs with various segmentation targets. MedSAM is substantially different than the two models described above. It does not fully automatically segment lungs in the CTs but rather requires the users to provide a prompt indicating where the lungs are. In this study, MedSAM processed 3D images as a series of 2D slices, with box prompts defined based on ground-truth lung masks.

For each model, its original code was used without modification for image preprocessing and lung segmentation to ensure reproducibility. Experiments were performed on an AMD EPYC 7302 (16-Core) with 512 GB of RAM and 1 NVIDIA A6000 GPU.

2.4 Evaluation Metrics

To evaluate automated lung segmentation models, we employed both quantitative and qualitative metrics.

Quantitative evaluation used three primary metricsVolumetric Similarity (versus), Dice Similarity Coefficient (DSC), and Hausdorff Distance (HD)—to objectively measure segmentation accuracy. These metrics were calculated between the predicted segmentation masks and the ground-truth masks in 3D for each case. Specifically, versus was calculated as a signed relative volume difference between the prediction and ground truth, which is unitless. A value of zero indicates perfect volume agreement, negative values indicate under-segmentation, and positive values indicate over-segmentation. For each metric, we reported the mean, standard deviation, and 95% confidence intervals (CIs) with 1000 bootstrapping, providing a comprehensive view of model performance. Detailed definitions and formulas for these metrics can be found in Appendix.

Qualitative evaluation (hereinafter referred to as clinical quality rate or CQR) incorporated expert radiologist assessment (M.R.H.) on 3D volume to ensure clinical relevance. Segmentation quality was rated based on clinical acceptability using a four-point scale:

- 1 (Poor): Errors account for a change in a single lung volume (left or right) of >15%, or if a pathology is completely or nearly completely missed by the segmentation.
- 2 (Acceptable): Errors account for a change in a single lung volume (left or right) of 5 – 15%.
- 3 (Good): Errors account for a change in a single lung volume (left or right) of <5%.
- 4 (Excellent): Fully acceptable with no clinical errors.

This qualitative assessment complements quantitative metrics by addressing clinically significant issues that may not be reflected in numerical scores, such as missing pathological features or including nonlung structures.

2.5 Statistical Analysis

Overall performance differences across models and within models were evaluated based on lung side, severity level, and pathology category, using statistical tests tailored to each comparison type. The ShapiroWilk[13] and the Levene's[14] tests were used to check normality and variance equality. For overall between-model comparisons on the same lung side or severity level, the Wilcoxon signed-rank test,[15] a non-parametric test for paired data, was used, with HolmBonferroni correction[16] applied to maintain strong Type I error control across multiple comparisons. Overall, within-model comparisons used the Wilcoxon signed-rank test for paired lung side comparisons and the MannWhitney U test[17] for independent severity level comparisons. For overall between-model comparisons on the same pathology type pairwise permutation test (10,000 permutations) with HolmBonferroni correction was applied due to the small sample size in some types. Overall within-comparisons among pathology types were assessed using independent permutation tests (10,000 permutations) with BenjaminiYekutieli correction[18] to

account for partial patient overlap across types. Pathology subgroup analysis was also performed: between-model comparisons used pairwise permutation tests with HolmBonferroni correction, while within-model comparisons relied on independent permutation tests. A two-tailed P-value of <0.05 was considered statistically significant. Statistical analyses were performed using Python (version 3.11) with SciPy (version 1.15) and statsmodels (version 0.14) libraries.

3 Results

This study included 32 patients (22 males, 10 females; mean age 60.3 ± 15.6 years) who underwent chest CT scans for various lung pathologies. The dataset encompassed a broad range of lung pathologies, with the most prevalent being atelectasis ($n = 24$), nodules ($n = 23$), and ground-glass opacities ($n = 19$). Less frequent pathologies included pneumothorax ($n = 4$) and masses ($n = 3$). Most patients ($n = 22$) had moderate-to-severe disease, while a smaller group ($n = 10$) exhibited only mild abnormalities. This diversity in pathology types and severity levels ensured a clinically representative cohort for segmentation model evaluation. Detailed data characteristics were presented in Table 2.

Table 2 Characteristics of the test dataset.

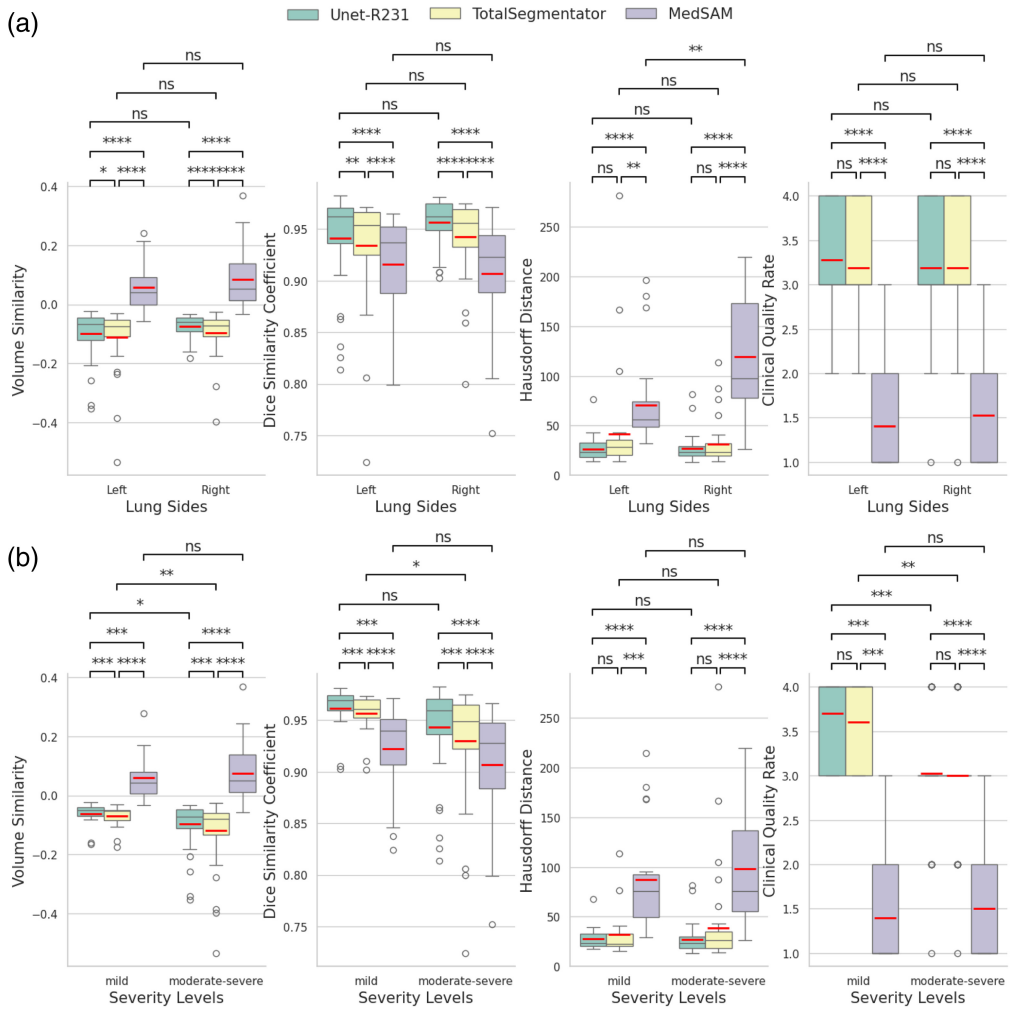
Variable	Dataset (no. of patients)
Age (y, mean ± SDs)	60.31 ± 15.58
Sex	
Female	10
Male	22
Race	
White/Caucasian	23
Black/African American	7
Multiracial	1
Not reported	1
Severity Levels	
Mild	10
Moderate-to-Severe	22
Pathologies (total, (mild: moderate-to-severe))	
Atelectasis	24 (21:3)
Consolidation	13 (8:5)
Emphysema	12 (5:7)
Fibrosis	10 (6:4)
Ground-glass opacities	19 (13:6)
Interlobular septal thickening	7 (0:7)
Mass	3 (3:0)
Nodule	23 (20:3)
Pleural effusion	16 (11:5)
Pleural thickening/nodularity	5 (0:5)
Pneumothorax	4 (1:3)

3.1 Overall Comparison Split by Lung Side

Based on Fig. 3(a), Unet-R231 consistently demonstrated the best performance, achieving the closest-to-zero versus highest DSC, lowest HD, and highest CQR. TotalSegmentator followed closely, with performance differences compared with Unet-R231 reaching significance for both lungs in versus and DSC ($p < 0.05$ and $p < 0.0001$ each), but not in HD or CQR. MedSAM lagged across all metrics with significantly lower accuracy and higher variability (mostly $p < 0.0001$). Within models, performance was slightly reduced in the left lung compared with the right, though this was only statistically significant for MedSAM in HD ($p < 0.05$).

3.2 Overall Comparison Based on Severity Level

As shown in Fig. 3(b), Unet-R231 remained the top performer and significantly outperformed TotalSegmentator in versus and DSC across both severity levels ($p < 0.001$); however, no significant differences were observed in HD or CQR between the two. All models showed decreased performance in moderate-to-severe cases compared with mild. These within-model declines were significant for Unet-R231 in versus ($p < 0.05$) and CQR ($p < 0.001$) and for TotalSegmentator in



Note-Panel A shows model performance by lung sides, and Panel B illustrates by severity levels. Each boxplot shows the distribution of model performance, with the red line indicating the mean, the box representing the interquartile range (IQR), the whiskers extending to 1.5 times the IQR, and individual points representing outliers. For between-model comparisons on the same lung side or severity level, statistical significance was tested using the Wilcoxon signed-rank test with Holm-Bonferroni correction. For within-model comparisons, it was tested using the Wilcoxon signed-rank test for lung sides and the Mann-Whitney U test for severity levels. Asterisks indicate significance levels (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001), and "ns" represents non-significant differences.

Fig. 3 Overall comparison of segmentation performance across models by lung side and severity level.

versus ($p < 0.01$), DSC ($p < 0.05$), and CQR ($p < 0.01$). MedSAM's performance was consistently poor and showed no significant differences between severity levels.

3.3 Overall Pathology Category Comparison

According to Fig. 4, UNet-R231 and TotalSegmentator maintained stable performance across most pathologies. Significant differences between these models emerged only in a few categories, such as fibrosis (versus $p < 0.01$) and four categories in DSC (e.g., fibrosis $p < 0.01$; ground-glass opacities, nodule, and pleural thickening/nodularity $p < 0.05$). HD and CQR showed no significant differences between the two. MedSAM consistently performed worse across nearly all pathologies and metrics ($p < 0.01$), except in mass and pneumothorax, where no between-model differences were detected. Within each model (Fig. 5), segmentation performance did not significantly vary across pathology types ($p > 0.05$).

3.4 Subgroup Lung Side Comparison

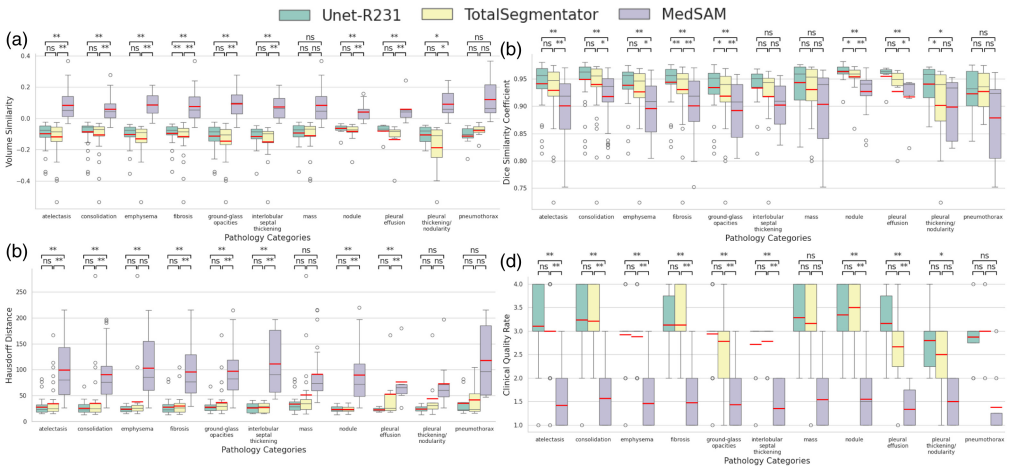
Based on Fig. 6, under most pathology categories, UNet-R231, TotalSegmentator, and MedSAM exhibited consistent segmentation performance between left and right lungs, with no significant side-to-side differences in versus DSC, HD, and CQR. Only MedSAM showed a significant underperformance on the right lung in HD ($p < 0.01$) for pneumothorax, despite its versus DSC and CQR remaining comparable across both sides.

3.5 Subgroup Severity Level Comparison

As shown in Fig. 7, models exhibited superior performance at a mild level compared with a moderate-to-severe level. At the mild severity level, Unet-R231 achieved significantly higher versus for atelectasis and consolidation, while TotalSegmentator outperformed for consolidation and fibrosis (all $p < 0.05$). For the DSC, both Unet-R231 and TotalSegmentator showed superior performance in consolidation ($p < 0.05$). In terms of HD, each model performed significantly better in fibrosis and ground-glass opacities (all $p < 0.01$). Finally, for the CQR, Unet-R231 significantly outperformed in consolidation ($p < 0.05$).

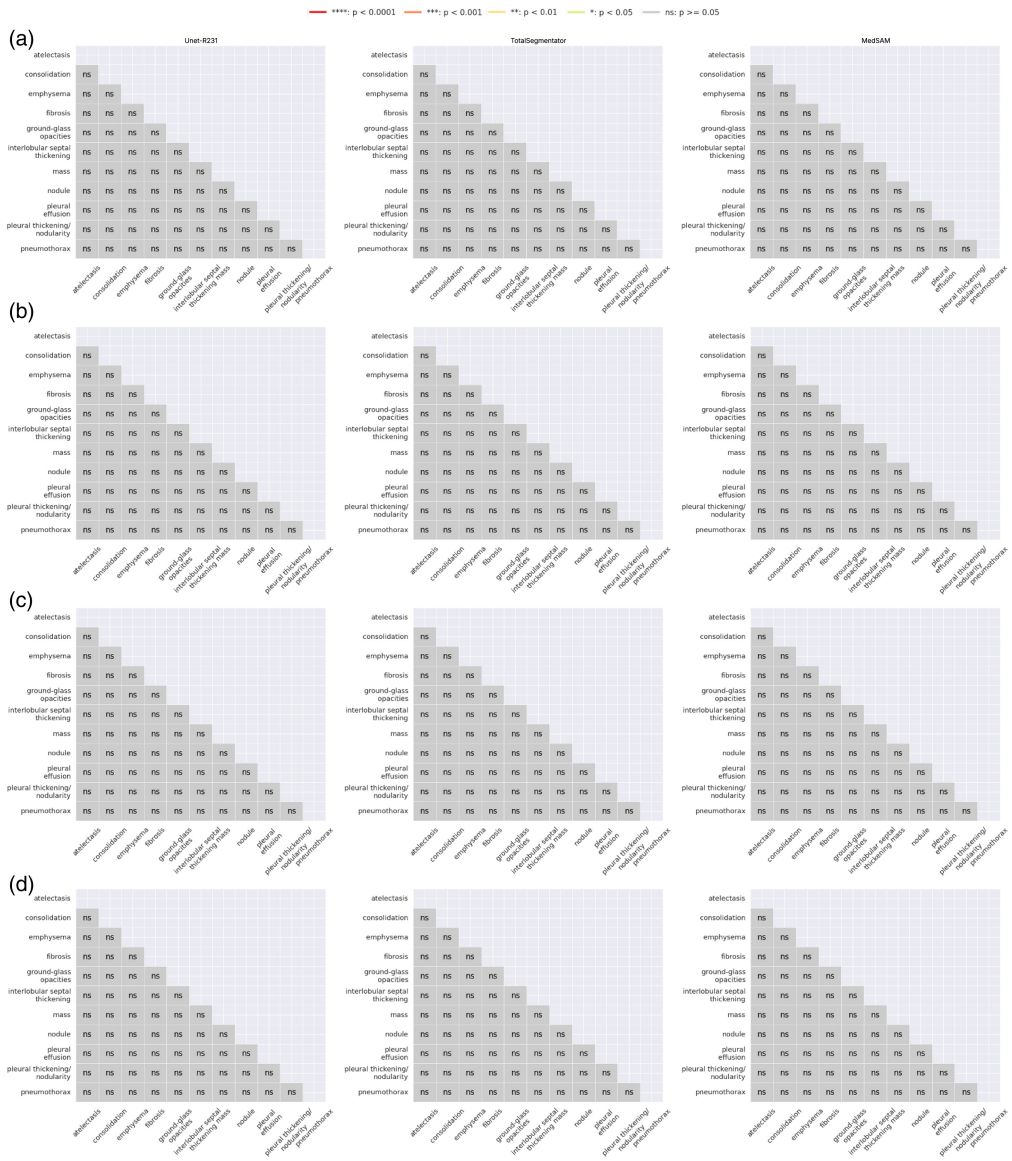
3.6 Sensitivity Analysis of MedSAM to Bounding Box Perturbation

To assess the robustness of MedSAM to variations in bounding box prompts, we further conducted a sensitivity analysis by perturbing the bounding boxes used at inference. In previous sections, bounding boxes were generated from reference lung masks with up to 10% random padding on each side. Starting from these baseline boxes, we applied two additional



Note-Each panel A, B, C, and D shows the distribution of model performance within a pathology category in each evaluation metric (in order of Volumetric Similarity, Dice Similarity Coefficient, Hausdorff Distance, and Clinical Quality Rate), with the red line indicating the mean, the box representing the interquartile range (IQR), the whiskers extending to 1.5 times the IQR, and individual points representing outliers. Statistical significance between models was tested using pairwise permutation tests (10,000 permutations) with Benjamini-Yekutieli correction. Asterisks indicate significance levels (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001), and "ns" represents non-significant differences.

Fig. 4 Overall between-model comparison of segmentation performance by pathology category.



Note-Each panel A, B, C, and D shows the distribution of model performance across pathology category in each evaluation metric (in order of Volumetric Similarity, Dice Similarity Coefficient, Hausdorff Distance, and Clinical Quality Rate). Statistical significance across pathology categories within a model was tested using independent permutation tests (10,000 permutations) with Benjamini-Yekutieli correction. Asterisks indicate significance levels (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001), and "ns" represents non-significant differences.

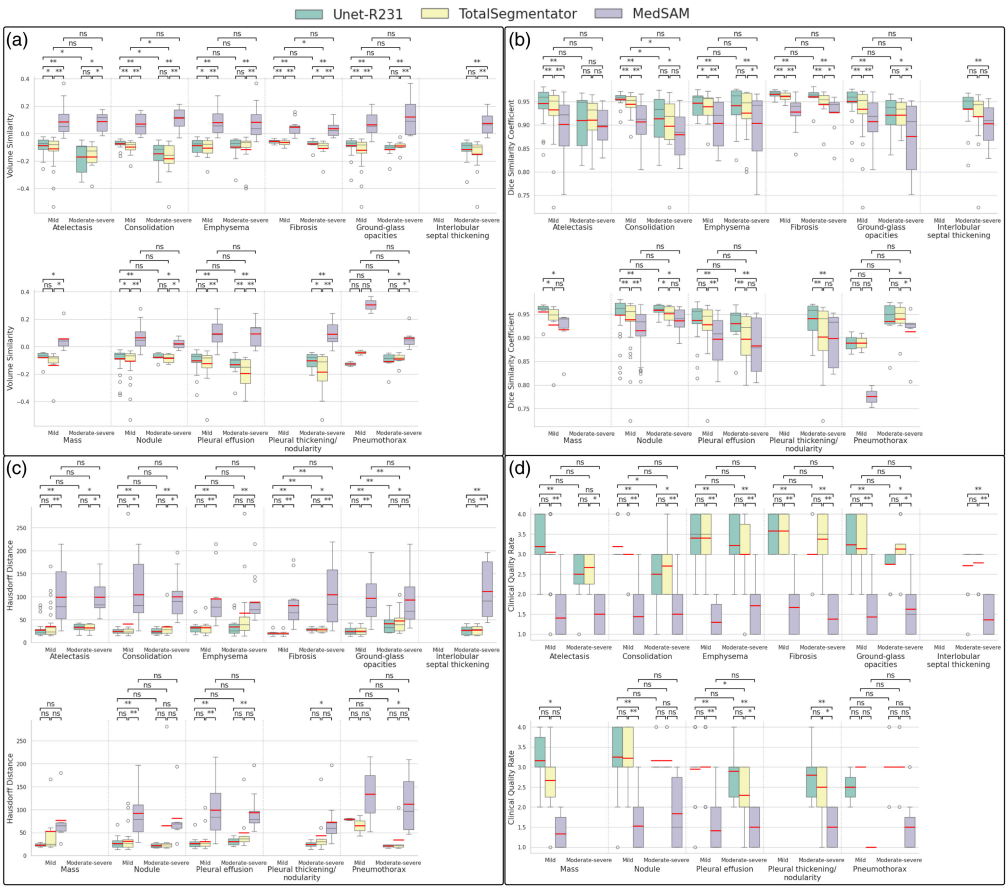
Fig. 5 Overall within-model comparison of segmentation performance by pathology category.

perturbations: 2% and 5% expansion with random shift. Specifically, each baseline bounding box was first expanded, and then randomly shifted within the expanded margin. These perturbed prompts were designed to simulate a more realistic user situation where the boxes are drawn more loosely and slightly misaligned with the lung region.

As shown in Table 3, MedSAM performance decreased as the bounding boxes became less precise. The Dice drops to 0.891 and 0.838 for the 2% and 5% expansion, respectively.

4 Discussion

This study evaluated three publicly available deep-learning-based models for automated lung segmentation in transplant-eligible patients. Our findings confirm that models developed for general populations do not consistently perform well in the pathologically distorted lungs commonly



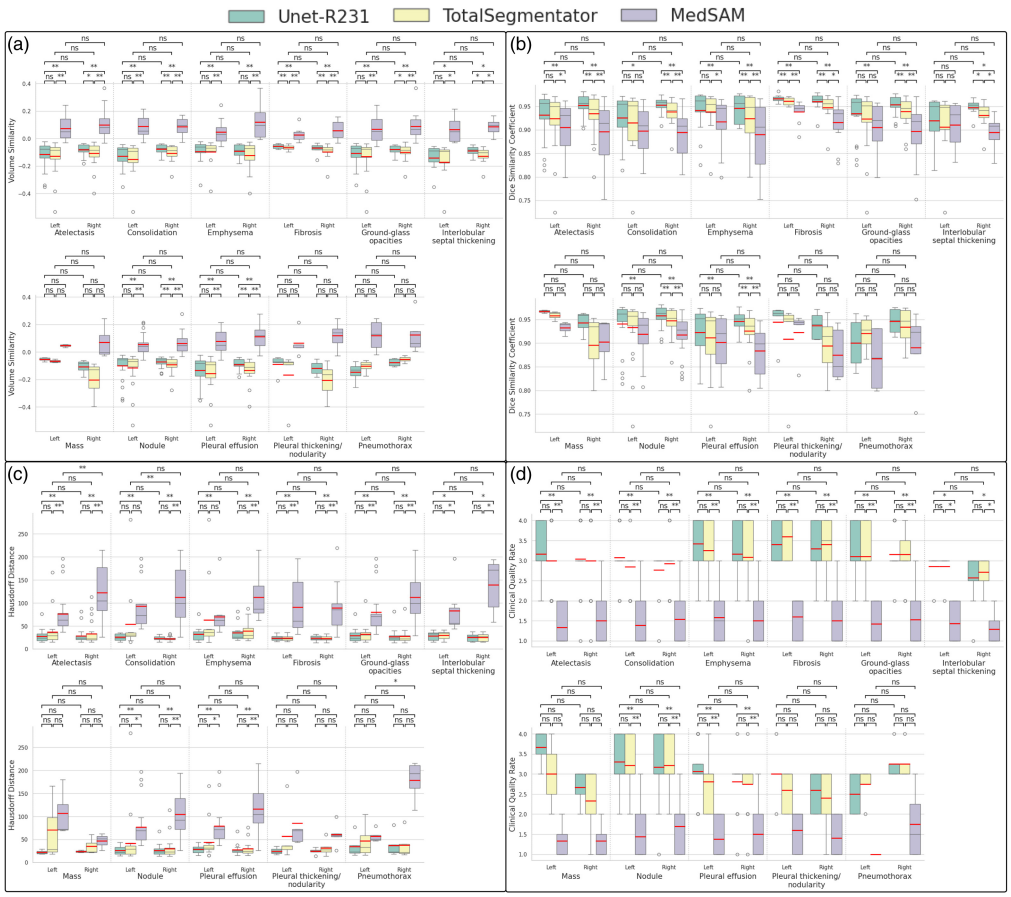
Note-Each panel A, B, C, and D shows the distribution of model performance within a pathology category in each evaluation metric (in order of Volumetric Similarity, Dice Similarity Coefficient, Hausdorff Distance, and Clinical Quality Rate), with the red line indicating the mean, the box representing the interquartile range (IQR), the whiskers extending to 1.5 times the IQR, and individual points representing outliers. Statistical significance was tested using pairwise (between-model comparison, 10,000 permutations) and independent (within-model, 10,000 permutations) permutation tests with Holm-Bonferroni correction. Asterisks indicate significance levels (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001), and "ns" represents non-significant differences.

Fig. 6 Pathology subgroup comparison of segmentation performance across models by severity levels.

seen in LTx candidates. Although Unet-R231 and TotalSegmentator showed relatively strong overall performance, both experienced notable performance declines in moderate-to-severe diseaseparticularly in versus and CQR, two metrics critical for preoperative planning. These results highlight a key limitation: even high-performing, publicly available models are vulnerable to segmentation errors in anatomically complex lungs.

Consistent with prior research, these findings highlight the importance of training dataset composition in determining model robustness. Unet-R231's superior performance likely stems from its training on chest CT datasets enriched with severe lung abnormalities, making it more adaptable to complex imaging appearances typical of LTx candidates. TotalSegmentator was trained on a broader set of whole-body CTs, with fewer severely diseased lungs, likely contributing to its slightly lower accuracy. In contrast, MedSAM's poor performance across all metrics reflects a fundamental limitation in both its training approach and model design, as it was developed using general medical datasets, which may dilute its capability on CTs, and relies on user-defined prompts rather than pathology-aware segmentation.

Importantly, this study also suggests that disease severity is an influential determinant of model reliability. While none of the models exhibited significant performance differences across specific pathology types, both Unet-R231 and TotalSegmentator showed consistent declines in segmentation accuracy with increasing disease severity. Given that most LTx candidates present with moderate-to-severe disease, these results emphasize the need for models that are optimized specifically for this subgroup to ensure reliable clinical application.



Note-Each panel A, B, C, and D shows the distribution of model performance within a pathology category in each evaluation metric (in order of Volumetric Similarity, Dice Similarity Coefficient, Hausdorff Distance, and Clinical Quality Rate), with the red line indicating the mean, the box representing the interquartile range (IQR), the whiskers extending to 1.5 times the IQR, and individual points representing outliers. Statistical significance was tested using pairwise (between-model comparison, 10,000 permutations) and independent (within-model, 10,000 permutations) permutation tests with Holm-Bonferroni correction. Asterisks indicate significance levels (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001), and "ns" represents non-significant differences.

Fig. 7 Pathology subgroup comparison of segmentation performance across models by lung sides.

Table 3 Sensitivity analysis of medSAM to bounding box perturbation.

Bounding box	Dice similarity coefficient	Volumetric similarity	Hausdorff distance
GT-based (original)	0.911 ± 0.051	0.079 ± 0.078	94.9 ± 55.2
2% expansion + shift	0.891 ± 0.055	0.132 ± 0.096	100.1 ± 51.6
5% expansion + shift	0.838 ± 0.062	0.246 ± 0.111	113.7 ± 51.5

In addition, a closer examination of pathology subgroups reveals that the impact of disease characteristics on performance varies considerably across pathology types and severity. As shown in Fig. 6, atelectasis and consolidation contributed the most substantial degradations in segmentation accuracy, probably due to the poorly defined boundaries between from the lung and surrounding tissues (as shown in Fig. 8). In contrast, fibrosis and nodules showed relatively less influence on model performance. Another finding is that pneumothorax cases showed equal or even improved performance despite their higher severity scores. One of the possible explanations is the smaller sample size (see Table 2)

Furthermore, our results suggest that deep-learning-based models can provide consistent segmentation across left and right lungs, supporting their use in side-specific transplant planning.

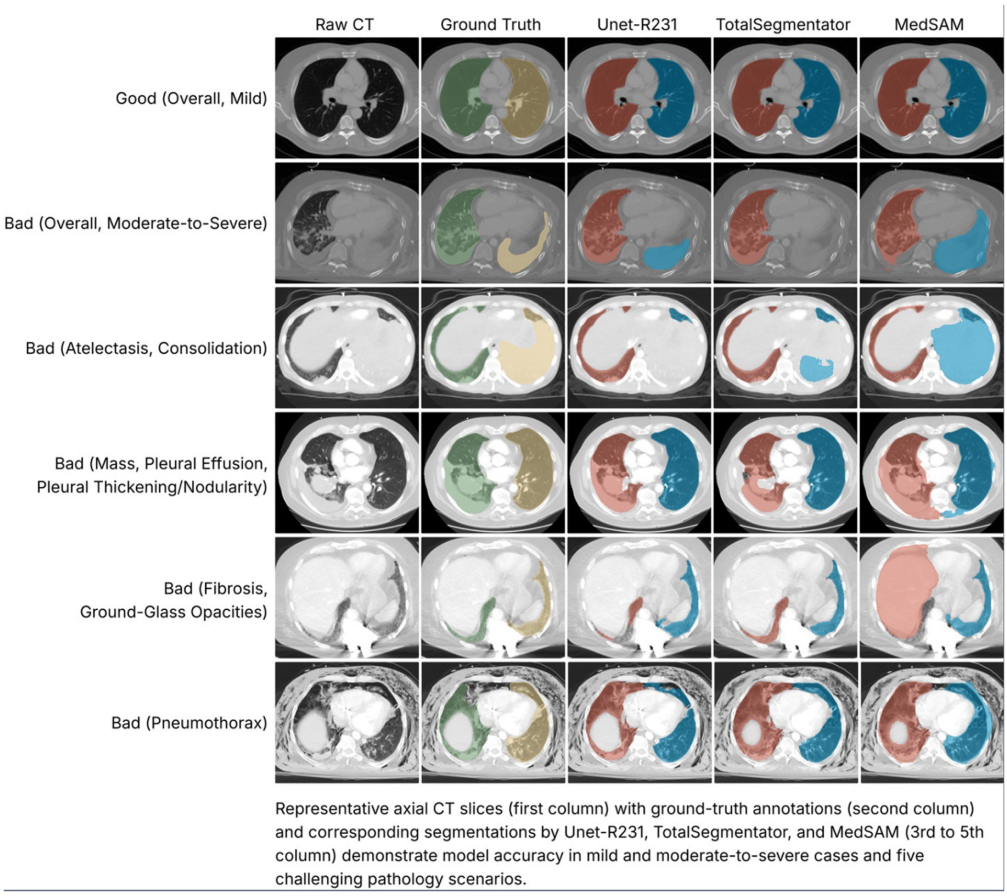


Fig. 8 Examples of lung segmentation model performance between severity levels and pathology categories.

However, both Unet-R231 and TotalSegmentator consistently produced slightly smaller segmentation volumes than ground truth (negative versus), likely due to the exclusion of peripheral and collapsed lung regions. This systematic under-segmentation could lead to underestimation of thoracic cavity volume and ultimately compromise donor-recipient size matchingan essential factor in transplant success. Therefore, we frame this study as an upstream validation step for LTx planning, as any CT-based volumetry analysis relies on robust lung segmentation in anatomically distorted lungs. From the clinical perspective, the observed negative versus suggests a trend to volume underestimation, and the larger performance declines in moderate-to-severe disease highlight cases where automated segmentation may benefit from additional quality review before being used for transplant planning.

This study has several limitations. First, the relatively small sample size (32 patients) and the imbalance in pathology type distribution may limit the generalizability of our findings. We addressed this by applying nonparametric tests and permutation methods to ensure robust statistical analyses. Second, all CTs were acquired at a single institution, but variations in imaging protocols and scanner types were not controlled, potentially impacting model performance. Third, potential confounders, such as prior surgical interventions or underlying comorbidities, were not explicitly accounted for, though the diverse pathology inclusion mitigates some bias. Fourth respiratory phase was not consistently available, and breathing-related volume variability may exceed the segmentation-related volume variations evaluated in this study. In addition, future work will compare CT-derived lung volumes with clinically accepted metrics (e.g., height-based equations/pTLC) and fine-tune models on LTx-specific data to improve robustness in moderate-to-severe disease. Moreover, MedSAM was evaluated using bounding boxes derived from ground-truth masks with random padding. The sensitivity analysis further demonstrated that MedSAM performance degraded progressively as bounding boxes became looser, as

presented in Table 3. This finding suggests that MedSAM is sensitive to prompt quality, and its performance may decline in practice when less precise bounding boxes are provided.

Despite these limitations, our findings offer significant clinical implications, particularly for lung transplantation and similar clinical contexts involving severe lung pathologies. First, future models should be trained or fine-tuned on datasets rich in moderate-to-severe pathologies to ensure robustness in complex cases. Second, standardized segmentation definitionsincluding clear guidelines on whether to include peripheral, non-aerated, or fluid-filled regionsare necessary to avoid underestimation in volumetric analyses.

5 Conclusion

This study evaluated three publicly available deep-learning-based lung segmentation models—Unet-R231, TotalSegmentator, and MedSAMon 32 transplant-eligible patients across varying lung pathologies, severity levels, and lung sides. Quantitative and qualitative assessments showed Unet-R231 performed best overall, but all models experienced significant accuracy declines in moderate-to-severe disease and exhibited systematic under-segmentation. These findings underscore the limitations of current models in complex pathological contexts and highlight the need for fine-tuning on severity-enriched datasets and establishing standardized segmentation protocols for reliable preoperative lung transplantation planning.

6 Appendix: Details on Quantitative Evaluation Metrics (Method)

Detailed definitions and formulas for quantitative evaluation metrics are provided here.

Volumetric Similarity (VS) quantifies the volume agreement between predicted (V_A) and ground truth (V_B) volumes: $\mathrm{VS} = 2(V_A - V_B)/(V_A + V_B)$. Negative versus indicates the predicted volume is underestimated, while positive versus denotes the predicted volume is overestimated. This metric is particularly valuable in preoperative lung assessment, where precise volume estimation is essential for transplant suitability.

Dice Similarity Coefficient (DSC) measures the spatial overlap between predicted (A) and ground truth (B) segmentations: $\mathrm{DSC} = \frac{2|A \cap B|}{|A| + |B|}$. A DSC of 1 indicates perfect overlap, while 0 denotes no overlap. This metric is widely used in medical image segmentation tasks.

Hausdorff Distance (HD) evaluates the boundary accuracy by calculating the maximum Euclidean distance ($D(a, b)$) between boundary pixels of predicted (A) and ground truth (B) segmentations: $H(A, B) = \max_{a \in A}\{\min_{b \in B} D(a, b)\}$. A lower HD indicates better boundary alignment, which is essential for cases involving lung abnormalities that require precise delineation for surgical planning.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code and Data Availability

Data and code generated or analyzed during the study may be available from the corresponding author by request.

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